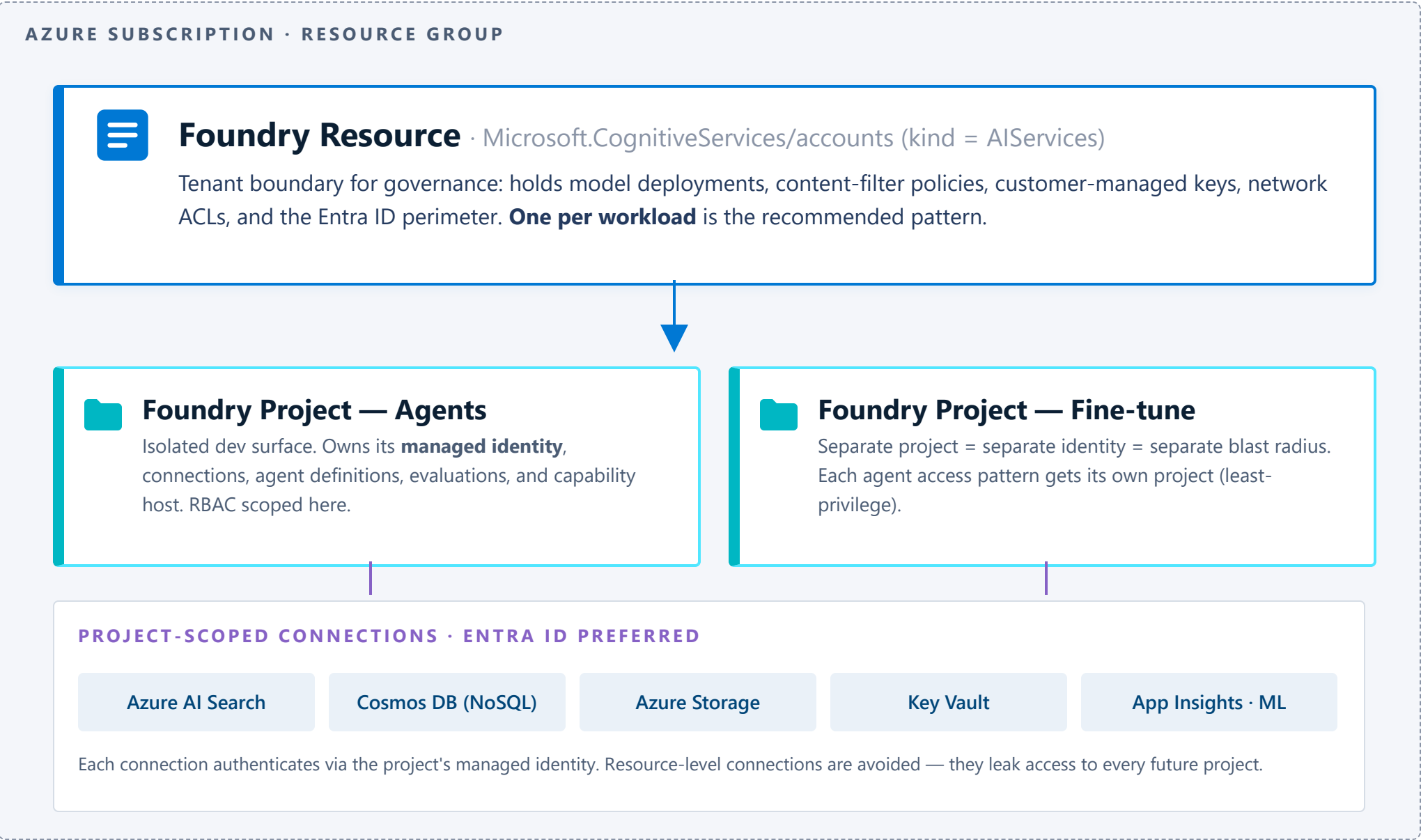


Resource Hierarchy & Control Plane

The three-tier resource model that anchors governance, identity, and project isolation across Foundry.



ARCHITECTURAL INVARIANTS

Why the hierarchy matters

- 01 Resource = Network & Identity Perimeter**

Private endpoints, customer-managed keys, content-filter policies, and Entra tenancy all bind to the Foundry resource — not the project.
- 02 Project = Least-Privilege Unit**

All agents in one project share one managed identity. Need distinct access? Split into separate projects — never elevate one identity.
- 03 BYO Dependencies, Service-Managed Data**

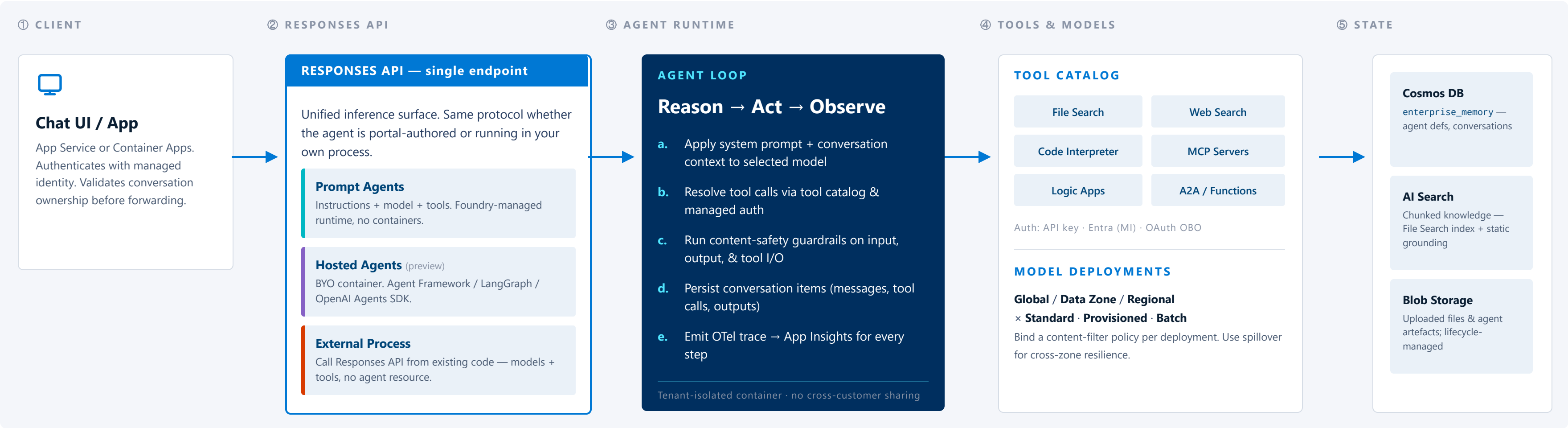
Agent state lives in your Cosmos DB `enterprise_memory`, files in your Storage, chunks in your AI Search — Foundry manages schema and data plane.
- 04 Hub-based Projects → Maintenance Mode**

New deployments use Foundry projects on the `CognitiveServices` RP; legacy ML hubs are sunset path.

DEEP ARCHITECTURE · PART 2 OF 3

Agent Service Runtime & Tool Execution

The Responses API as the single entry point — agent loop, tool orchestration, and stateful conversation persistence.



STATEFUL BY DEFAULT

Conversation IDs persist multi-turn context across sessions — but the app server, **not Foundry**, must enforce per-user ownership (BOLA).

IDENTITY-FIRST AUTH

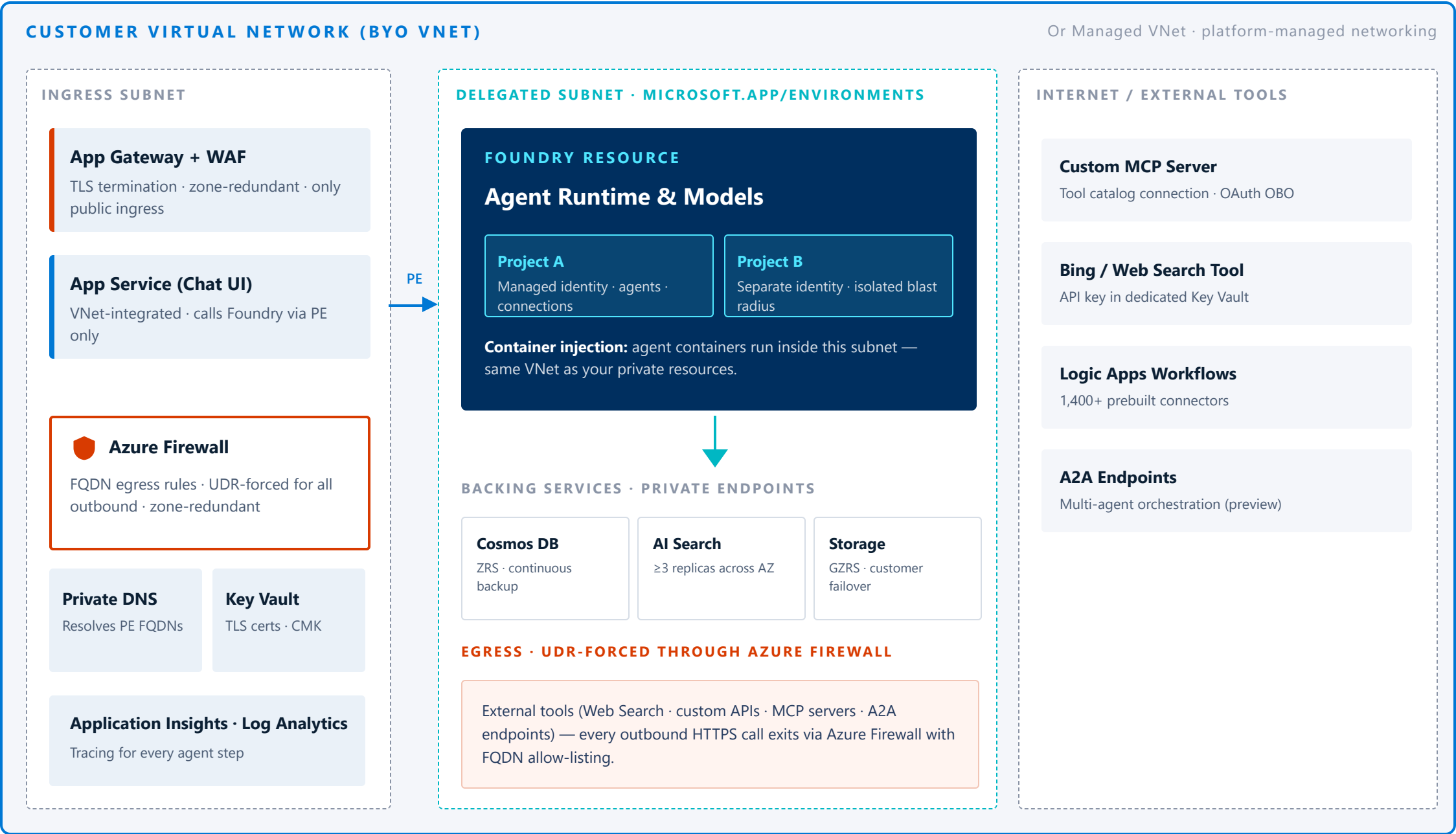
Each agent can have a dedicated Entra ID. Tools reach downstream APIs via managed identity or OAuth On-Behalf-Of — no shared keys.

OBSERVABLE END-TO-END

Every model call, tool invocation & decision is traced. Evaluations + agent optimizer close the regression loop.

Network Isolation & Enterprise Security

Private endpoints for ingress · subnet-delegated egress · Azure Firewall as the egress chokepoint.



DEFENSE IN DEPTH

Five enforcement layers

- 1 Network — Private Link everywhere**
All PaaS traffic over Azure backbone. Foundry, Cosmos, AI Search, Storage, Key Vault reached only via private endpoints.
- 2 Identity — Entra-first, MI per agent**
Separate identities for resource, project, web app, agents. Tool auth via managed identity or OBO — no shared secrets.
- 3 Content — Guardrails & XPIA shields**
Filters at user input, model output, tool calls and tool responses. Cross-prompt injection mitigation built into runtime.
- 4 Data — Bring your own, manage lifecycle**
Cosmos / Storage / AI Search in your subscription. Apply CMK, delete-locks, continuous backup, ZRS/GZRS as you would any data tier.
- 5 Governance — Agent-as-code, IaC only**
Block portal-based project creation in prod (it bypasses your network controls). Define agents, prompts and connections in source control.